

SHIKSHIT PARAJULI

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July, 2026

EDUCATION

University of Georgia

MS in Agricultural and Applied Economics

GPA: 4.0/4.0

Athens, GA

Expected Aug 2026

Agriculture and Forestry University

MSc in Agricultural Extension

CGPA: 3.41/4.0

Chitwan, Nepal

Feb 2024

BSc in Agriculture

CGPA: 3.55/4.0

2020

TECHNICAL PROJECTS: REMOTE SENSING & AGRICULTURAL INDEX INSURANCE

• Rainfall Index & NDVI Co-Movement Analysis

R, Google Earth Engine, panel econometrics

- Built a full panel dataset (12 Georgia counties $\times$ 6 years) directly from satellite and climate data using Google Earth Engine: Sentinel-2 imagery (NDVI via band math and cloud masking) and PRISM precipitation.
- Estimated a two-way fixed-effects regression testing whether USDA RMA's real precipitation-based Rainfall Index co-moves with satellite-measured vegetation health, with county-clustered standard errors (*R*, *fixest*).
- Result consistent with published literature on the same USDA program (Keeler & Saitone; Ballagh et al. 2012), directly speaking to basis risk in agricultural index insurance design.

• NDVI & Cotton Yield: STAX/ARPI Analysis

R, Google Earth Engine, USDA NASS data

- Companion analysis re-targeting the NDVI pipeline above at insurance products that actually apply to row crops (STAX, Area Risk Protection Insurance), using real NASS Quick Stats county-level cotton yield data.
- Constructed an Area Yield Index mirroring STAX/ARPI's actual payout structure and found a significant relationship between satellite vegetation health and realized yield outcomes.

• Index Insurance and Basis Risk: A GIS Illustration

ArcGIS Online, ArcGIS StoryMaps

- Built an interactive GIS narrative showing how a computed rainfall index diverged sharply across three neighboring Georgia counties in a single growing season, illustrating basis risk in agricultural index insurance, plus a 2017–2026 land cover time series for longer-term risk signals.

• Basis Risk in Cotton Insurance: NDVI vs. STAX/ARPI

ArcGIS Online, ArcGIS StoryMaps

- Built an interactive GIS narrative testing whether satellite-measured vegetation health (NDVI) tracks the Area Yield Index used by USDA's STAX and ARPI programs to trigger cotton insurance payouts, using real Sentinel-2 imagery and NASS yield data across 12 Georgia counties to visualize basis risk for a general audience.

• Cotton Acreage Forecasting

Python, pandas, statsmodels

- Rebuilt a published cotton acreage forecasting model (Southern Ag Today, 2025) in Python: OLS regression with Newey-West standard errors, Augmented Dickey-Fuller testing, and a bootstrap Monte Carlo simulation for stochastic acreage projection using real NASS/WASDE data.

• Cover Crop Economics Dashboard



R, Shiny

- Interactive decision-support tool comparing cost, yield, and net returns across cover crop management systems under user-defined price scenarios.

HONORS & AWARDS

Outstanding MS Student Award

2026

Dept. of Agricultural and Applied Economics, University of Georgia

RESEARCH EXPERIENCE

• Graduate Research Assistant

August 2024 - Present

Department of Agricultural and Applied Economics, University of Georgia

- Analyze long-term experiment and farm-level data to evaluate the profitability of adopting various cover crop alternatives in Georgia row crop systems.
- Implement econometric and risk-analysis methods including Stochastic Efficiency with Respect to a Function (SERF), using R, STATA, and SIMETAR to rank alternative cover crop strategies, determine risk premium, and run sensitivity analysis.
- Clean, merge, and document large, multi-source datasets supporting ongoing research on conservation practice adoption and farm-level decision making.
- Contribute to manuscript preparation, conference presentations, and extension outputs on cover crops, risk management, and Georgia cotton and row crop production.
- Conducted cotton acreage forecasting using historical acreage and price data, combining time-series analysis with Monte Carlo simulation to generate stochastic acreage projections for Georgia.

MS Thesis

(2026)

Department of Agricultural and Applied Economics, University of Georgia

Title: "Economic and Risk Implications of Cover Crop Adoption in Georgia Row Crop Systems"

Major Advisor: Dr. Yangxuan Liu

Member Advisors: Dr. Gregory Colson, Dr. Wesley Porter

MSc Thesis

2023

Department of Agricultural Extension and Rural Sociology, Agriculture and Forestry University

Title: "A study on entrepreneurial intention among undergraduate agriculture students of Nepal"

Major Advisor: Dr. Ram Hari Timilsina

PEER-REVIEWED PUBLICATIONS

• Journal Publications

- Subedi, A. P., Jaishi M., & **Parajuli, S.** (2025). "Analyzing Extension and Advisory Service Delivery Dynamics in Nepal through Social Network Analysis Lens: A Comparative Study". *Asian Journal of Agricultural Extension, Economics & Sociology* 43(1):124-42. <https://doi.org/10.9734/ajaees/2025/v43i12679>
- **Parajuli, S.**, & Thapa, R. (2024). "Factors affecting the technical efficiency of potato production under the command area of the PMAMP project in Nepal." *Cogent Food & Agriculture*, 10(1). <https://doi.org/10.1080/23311932.2024.23607549>
- Bista, B., Dhital, P.R., Acharya, S., Dahal, B., & **Parajuli, S.** (2021). "Gender participation in activities and decisions of vegetable cultivation in Surkhet, Nepal." *Food & Agribusiness Management*, 2(1): 10-13. <https://doi.org/10.26480/fabm.01.2021.10.1>

• Extension Publications

- **Parajuli, S.**, & Liu, Y. (2025). "Projections for 2025 Cotton Acreage." *Southern Ag Today*, 5(18.3). <https://southernagtoday.org/2025/04/30/projections-for-2025-cotton-acreage-2/>

TEACHING EXPERIENCE

• Graduate Teaching Assistant

- Agribusiness Financial Management (AAEC 6740E), Fall 2025
Instructor: Dr. Yangxuan Liu — financial statement analysis, DuPont credit analysis, capital budgeting.
- Farm Organization and Management (AAEC 3010), Fall 2024
Instructor: Dr. William Secor — farm finance, risk appraisal, investment analysis.
- Hospitality Beverage Management (HFIM 4090), Spring 2025
Instructor: Dr. Daniel Remar.

PROFESSIONAL EXPERIENCE

Agricultural Officer

2022-2024

Malika Rural Municipality, Myagdi, Nepal

- Served as focal person for the Farmers' Management Information System (MIS), enrolling nearly 80% of agricultural farms and improving data availability for local planning.
- Led preparation of the "Rural Revitalization Project" (approx. USD 1.4 million), advancing to the top 5 of 37 applicants in a competitive provincial funding program.
- Facilitated planning, implementation, and reporting of annual agricultural programs for a rural municipality of nearly 2,000 agriculture-dependent households.

PRESENTATIONS

(* indicates presenter among co-authors)

Parajuli, S. (*), Bobbie, K., Liu, Y., Porter, W., Reagin, K., Tostenson, M., Sapp, P., Mallard, J., and McAllister, S. "Economic and Risk Implications of Cover Crop Adoption in Georgia Row Crop Systems". *Southern Agricultural Economics Association Annual Meeting*. Selected paper. Louisville, KY. Feb 3, 2026.

Parajuli, S. (*), Bobbie, K., Liu, Y., Porter, W., Reagin, K., Tostenson, M., Sapp, P., Mallard, J., and McAllister, S. "Assessing the profitability of cover crops in Georgia row crop farming". *Beltwide Cotton Conference*. San Antonio, TX. Jan 8, 2026.

RELEVANT COURSEWORK

Microeconomics I & II, Applied Econometrics, Quantitative Techniques in Agricultural Economics, International Trade, Land Economics, Futures and Options Markets, Modern Statistical Programming

SKILLS

Programming & Tools: R (tidyverse, fixest, Shiny, ggplot2), Python (pandas, statsmodels, numpy), STATA, SQL, Git/GitHub, L^AT_EX

Geospatial & Remote Sensing: Google Earth Engine, ArcGIS Online, ArcGIS StoryMaps, Sentinel-2/satellite imagery analysis

Methods: Panel/fixed-effects econometrics, time-series forecasting, Monte Carlo simulation, SERF risk analysis, SIMETAR

Languages: Nepali (Native), English (Proficient), Hindi (Proficient)

GRANTS AND PROJECTS

Parajuli, S. (2023). "*Entrepreneurial Intention among agriculture students of Nepal*". DEPROSC Nepal Research grant, NPR 101,000 (approx. USD 757), 2023-24.

Parajuli, S. (2022). "*Rural Revitalization Project*", Provincial and Local Governance Support Program – Innovative Partnership Fund, Gandaki Province, Nepal.

MEMBERSHIP

Southern Agricultural Economics Association

2024-Present